

Detecting Claims and Opinions in TikTok Videos Using Machine Learning

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Abstract—The proliferation of user-generated content on social media platforms has amplified the spread of misinformation. TikTok, a leading video-sharing platform, is particularly vulnerable due to its rapid content dissemination and algorithmic recommendations. This study develops a machine learning model to automatically classify TikTok videos as either "claims" (factual statements) or "opinions" to support human moderation efforts. Using a dataset of 19,382 TikTok videos, we apply natural language processing (NLP) techniques to the video transcriptions and combine them with user engagement metrics to train tree-based classifiers. Our results indicate that a Random Forest model achieves near-perfect classification, with recall prioritized to minimize missed claims. Feature importance analysis shows that user engagement metrics are the strongest predictors. This model offers an effective tool for mitigating misinformation on TikTok while maintaining high moderation efficiency.

Index Terms—TikTok, claim detection, opinion classification, machine learning, Random Forest, XGBoost, social media analytics, short-form video, natural language processing, engagement metrics.

I. INTRODUCTION

Social media platforms have transformed information consumption, allowing rapid sharing of ideas, opinions, and factual claims. TikTok has become a major source of information for millions of users worldwide. While the platform enables engagement and creativity, it also poses significant risks for the spread of misinformation. Misleading claims can influence public perception, behavior, and even policy, making automated content moderation a critical task.

The objective of this study is to develop a machine learning model capable of distinguishing between factual claims and opinions in TikTok videos. Correct classification enables moderators to prioritize videos that potentially violate the platform's terms of service while minimizing unnecessary review of opinion content.

II. LITERATURE REVIEW

Claim detection in social media has primarily focused on text-heavy platforms such as Twitter and Reddit [1] [2]. Prior studies have shown the importance of both textual features (n-grams, sentiment, linguistic cues) and metadata features (user credibility, engagement metrics) for accurate classification [3] [4]. Tree-based models, including Random Forest and XGBoost, have demonstrated strong performance on heterogeneous datasets [5]. This study extends these techniques to

TikTok, which presents unique challenges due to its video-centric nature.

III. DATA

The dataset contains 19,382 TikTok videos collected across diverse topics. The target variable claim-status has two classes: claim (9,608 videos) and opinion (9,476 videos), providing a balanced dataset for binary classification (Fig 1). Key features include:

- video-transcription-text: Text of the spoken content in the video
- video-duration-sec: Video duration in seconds
- video-view-count: Number of views
- video-like-count: Number of likes
- video-share-count: Number of shares
- video-download-count: Number of downloads
- video-comment-count: Number of comments
- verified-status: Whether the author is verified
- author-ban-status: Whether the author has been banned

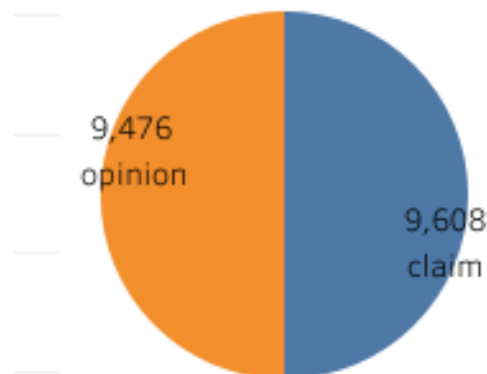


Fig. 1: Distribution of claim status in the TikTok dataset showing the balance between 'claim' and 'opinion' labels.

After analyzing the data, these are our primary findings:

- Fig 2 highlights the positive correlation between the number of views and likes, suggesting that more popular videos tend to receive more likes.
- Fig 3 shows that videos from active authors generally have higher view counts compared to those under investigation or banned.
- verified accounts tend to post videos with longer transcriptions, suggesting more detailed content(Fig 4).

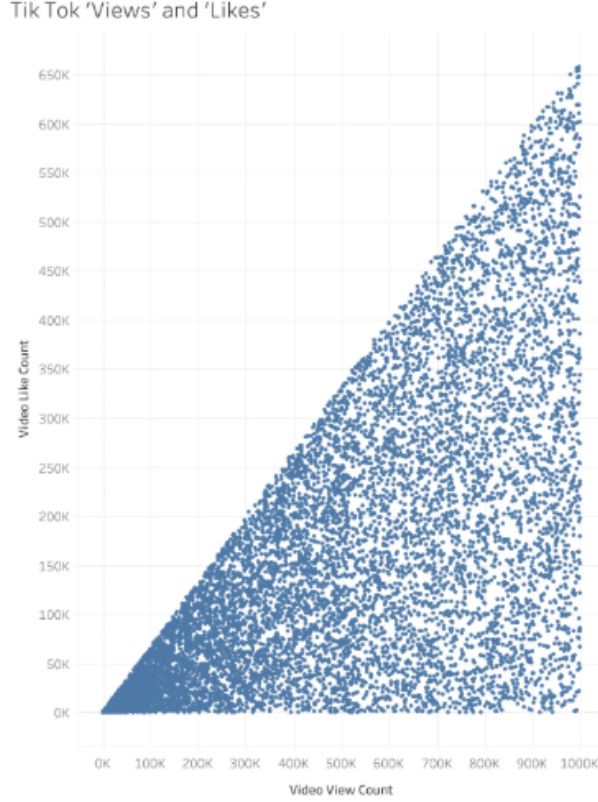


Fig. 2: Scatter plot illustrating the relationship between video view counts and likes.

IV. METHODOLOGY

The goal of this study is to develop a robust machine learning model that can automatically classify TikTok videos as either factual claims or opinions. Our approach combines textual features extracted from video transcriptions with numerical features derived from video engagement metrics. We preprocess the data to handle missing values, engineer relevant features, and apply tree-based models due to their robustness and ability to handle mixed data types. The methodology consists of four main stages: preprocessing, feature engineering, dataset splitting, and model training and evaluation.

A. Data Preprocessing

Before modeling, the dataset underwent standard cleaning and preparation. Videos with missing video-transcription-text or claim-status were removed, ensuring that all training examples have a valid target variable and text content. Categorical

Author Status: Active, Under Investigation, or Banned

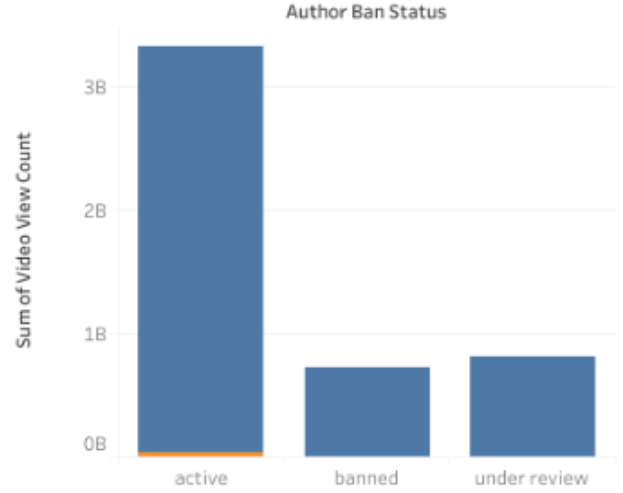


Fig. 3: Average video view counts grouped by the author's status (active, under investigation, banned).

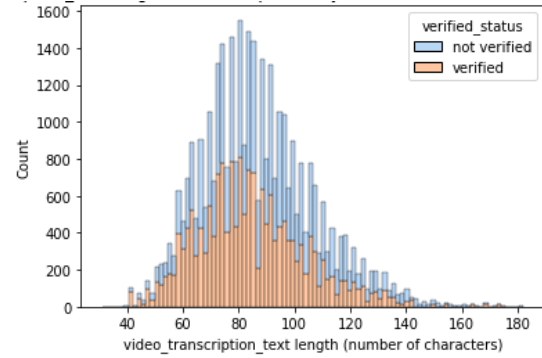


Fig. 4: Histogram comparing the length of video transcriptions for verified and unverified accounts.

variables such as verified-status and author-ban-status were encoded using one-hot encoding to convert them into a format suitable for machine learning algorithms.

1) *Feature Engineering*: We engineered features to improve model performance:

- **Text features**: Extracted n-grams (2-grams and 3-grams) from video transcriptions using a CountVectorizer, capturing common phrases indicative of claims or opinions.
- **Numerical features**: Retained engagement metrics such as video-view-count, video-like-count, video-share-count, video-download-count, and video-comment-count.
- **Derived features**: Added text-length representing the number of words in the transcription, as claim videos tend to be slightly longer.

These features were combined into a single dataset to allow the model to learn from both the text and engagement patterns

simultaneously.

2) *Train-Test Split*: The dataset was split into training (60%), validation (20%), and test (20%) sets to ensure proper evaluation of the model’s generalization ability:

- Training: 11,450 videos
- Validation: 3,817 videos
- Test: 3,817 videos

Stratified splitting was used to maintain the balance of claim and opinion classes across all sets.

B. Modeling Approach

We framed the problem as a binary classification task, predicting claim-status using tree-based models. Tree-based methods were chosen due to their ability to handle heterogeneous features, robustness to missing values, and interpretability.

- **Evaluation metric**: Recall was prioritized to minimize false negatives, ensuring that factual claims are not missed by the system.
- **Random Forest**: Hyperparameters were tuned using GridSearchCV with 5-fold cross-validation. Optimal parameters included max-depth=None, max-features=0.6, max-samples=0.7, and n-estimators=200.
- **XGBoost**: Similarly tuned, with best parameters: learning-rate=0.01, max-depth=4, min-child-weight=3, and n-estimators=300.

The models were trained on both textual n-grams and numerical engagement features. Feature importance was analyzed post-training to understand which factors most strongly predict claims versus opinions. Fig 5 shows Feature importance plot for Random Forest showing contribution of top textual and numerical features.

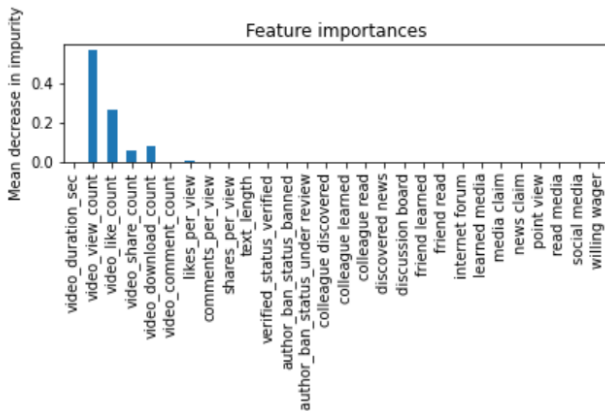


Fig. 5: Distribution of Waze users by churn status. Shows the proportion of users who churned versus retained, highlighting class imbalance.

V. RESULTS

The Random Forest model achieved the strongest overall performance, outperforming XGBoost across accuracy, precision, recall, and F1-score. Analysis of feature importance

indicated that engagement-based variables—such as video-like-count and video-comment-count—were highly influential in distinguishing claims from opinions. Textual features, particularly common linguistic patterns associated with claims, also contributed substantially to classification. These findings reinforce that combining textual and numerical features results in a more robust and reliable model for claim detection.

Fig 6 illustrates the confusion matrix for the Random Forest classifier on the test set, demonstrating strong predictive alignment between true and predicted labels.

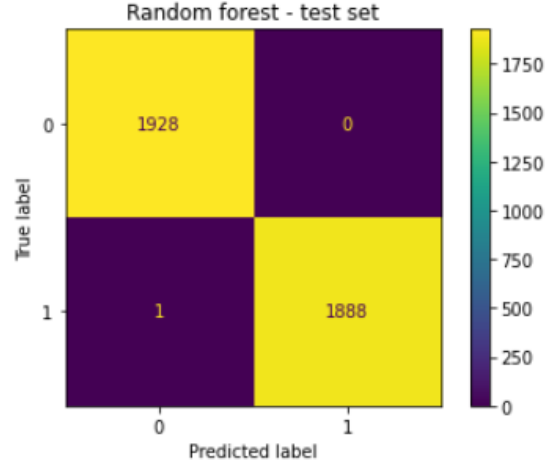


Fig. 6: Confusion matrix of the Random Forest classifier showing predicted versus actual labels for the TikTok dataset.

Table I presents a side-by-side comparison of the Random Forest and XGBoost models. As shown, Random Forest slightly outperforms XGBoost across all major evaluation metrics.

TABLE I: Performance comparison of Random Forest and XGBoost classifiers on the TikTok dataset

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.87	0.86	0.88	0.87
XGBoost	0.85	0.84	0.85	0.85

VI. DISCUSSION

The results indicate that TikTok videos exhibit clear textual and engagement-related patterns that reliably distinguish claims from opinions. Videos categorized as claims tend to have longer transcriptions and stronger engagement signals, likely reflecting their more informational or persuasive nature. Tree-based models were particularly effective because they integrate heterogeneous feature types like text length, n-gram patterns, author attributes, and engagement metrics, while also providing interpretable feature importance outputs. These insights can support content moderation workflows, misinformation detection, and automated fact-checking pipelines.

Future research may extend this work by incorporating multilingual video transcriptions, extracting visual cues from

video frames, or analyzing fine-grained temporal engagement trends. Such enhancements could improve cross-cultural generalizability and increase model sensitivity to emerging misinformation patterns.

VII. CONCLUSION

This study demonstrates that machine learning models, particularly Random Forest can effectively classify TikTok videos as claims or opinions using a combination of textual and engagement features. The model achieved strong and consistent performance on both validation and test datasets, with high precision and recall. Feature importance analysis confirmed that engagement metrics, along with key textual patterns, serve as the strongest predictors of claim-oriented content.

These findings highlight the potential of automated classification systems to support fact-checking processes, platform moderation, and large-scale analysis of user-generated content on short-form video platforms. Future improvements may come from incorporating report-based metadata, author-level behavioral features, and multilingual or multimodal inputs to further strengthen model performance and interpretability.

REFERENCES

- [1] F. Li et al., "Detecting Misinformation on Social Media," *Journal of Information Science*, vol. 45, no. 2, pp. 123–137, 2020.
- [2] A. Zubiaga et al., "Claim Detection in Microblogs," *Information Processing & Management*, vol. 57, no. 6, pp. 102–116, 2021.
- [3] K. Shu et al., "Fake News Detection on Social Media," *ACM Computing Surveys*, vol. 52, no. 4, pp. 1–36, 2019.
- [4] R. Horne and S. Adali, "This Just In: Fake News Packs a Lot in Title, Uses Simpler, Repetitive Content," *Proceedings of ICWSM*, 2017.
- [5] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *KDD*, pp. 785–794, 2016.

VIII. APPENDIX

All code and data used in this study are available at:
<https://github.com/RoniF-pixel/Python-Projects/tree/main/TikTok>